

TDS Proxmox Host — RAM Upgrade Spec

Dell PowerEdge R540-class (board 04JN2K) · single Xeon Silver 4110 · prepared 2026-06-07

Current State

CPU	1× Intel Xeon Silver 4110 (Skylake-SP, 8c/16t) — mem controller caps at DDR4-2400
Installed RAM	1× 8 GB Micron DDR4-2666 ECC RDIMM (1Rx8, PN 9ASF1G72PZ-2G6J1) in slot A1
Slots	16 total · 15 empty · single CPU = only CPU1 channels usable now
Problem	Running single-channel (worst bandwidth); 8 GB total bottlenecks everything
Board max	1 TB (768 GB per CPU)

Module Spec (must match for this CPU/board)

- **Type:** DDR4 **ECC Registered (RDIMM)** — not UDIMM, not LRDIMM
- **Speed:** PC4-2666V (2666 MT/s) — runs at 2400 on the 4110; same used price as 2400, so buy 2666
- **Rank:** 2Rx8 or 2Rx4 (dual-rank) for better bandwidth
- **Do NOT buy** 2933 / 3200 (2nd/3rd-gen Scalable spec) — downclocks anyway, wasted money
- **Channels:** 6 per CPU — populate in sets of 6 for full bandwidth

Recommended Configs

Config	Kit	Used price (2026)	Notes
96 GB ★	6× 16 GB 2Rx8 PC4-2666 RDIMM	~\$80–110	RECOMMENDED — full 6-channel, runs everything + small I/O
192 GB	6× 32 GB 2Rx4 PC4-2666 RDIMM	~\$180–240	Headroom for a real local model (30B-class quantized)
64 GB	4× 16 GB 2Rx8 PC4-2666 RDIMM	~\$55–75	Budget — only 4 of 6 channels; 96 GB barely costs more

★ The price/performance knee. Pull the lone 8 GB stick and run a matched 6-DIMM set.

Install Notes (specific to this box)

- **Remove the existing 8 GB stick** — don't mix it with 16 GB dual-rank (mismatched ranks hurt interleaving + throws config warnings).
- **Populate per Dell's DIMM matrix** — white slots first, one per channel on CPU1, for 6-channel interleave. Order is silkscreened on the board and in the R540 manual; iDRAC shows the map.
- **Single CPU = CPU1 slots only.** The other bank is dead until a 2nd CPU is added. 6× DIMMs fits CPU1's six channels.
- **Non-Dell RAM works** (14G isn't hard-locked) but iDRAC may log a cosmetic 'non-validated memory' SEL warning — harmless. Dell-branded pulls avoid it for a few dollars more.
- **Cold boot after install** — this is the box with the PCIe ASPM warm-reboot hang; cold boots are clean. Power down fully, seat DIMMs, cold start.
- **Verify after:** 'dmidecode -t memory', confirm Proxmox sees the full amount, check iDRAC SEL for DIMM training errors.

Shopping search strings (paste into eBay / Newegg)

- > 16GB 2Rx8 PC4-2666V ECC RDIMM (for 96 GB: buy qty 6)
- > 16GB DDR4-2666 ECC Registered server memory
- > 32GB 2Rx4 PC4-2666V ECC RDIMM (for 192 GB: buy qty 6)
- > Dell 14G R540 16GB DDR4-2666 RDIMM (Dell-branded, avoids SEL warning)
- > Example Dell PN to search: SNPHNDJ7C/16G (16GB) · SNP8WKDYC/32G (32GB)